



李明东

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About Me

我拥有 机械工程 和 计算神经科学 求学背景；我的研究兴趣在于 **Brain-machine Interface, Computational Neuroscience, Robotics**；我的博士方向是信号点过程滤波（point process filtering）分析神经群体编码和脑机接口系统；我的工作主要负责数采工厂数据管线建设、多模态操作算法和基座模型预训练。

教育经历

- 2021.09 – 2025.10 香港科技大学, PhD in Robotics and Autonomous Systems
Thesis: Modeling Functional Neural Connectivity in Point Process Filtering for Brain-machine Interface
导师: 王怡雯、陈启峰 (co-supervisor)
- 2018.09 – 2021.04 浙江大学, 学术硕士（机械设计及理论）
导师: 冯毅雄（长江学者，谭建荣院士团队）
- 2013.09 – 2018.07 同济大学, 本科（机械设计制造及其自动化，试验班 5 年制）

工作经历

1. 安克创新科技股份有限公司 (Humanoid Lab, 灵巧操作算法工程师) 2025.04 – 至今
- 负责 UMI 多模态、基座模型预训练团队。重点推进语言、触觉在前期 VA 模型探索基础下的集成；基于 DiT 探索视觉、语言、触觉模态的融合方案；主导多任务操作模型实现、基座操作模型预训练架构设计；探索多源 UMI 数据的预训练架构和 WAM 的应用。
 - 主导自研 UMI 数据采集体系设计与训练落地，包括数采工厂基础设施建设、数据格式定义、对接数据管线团队推进数采工人培训和数据质检生产。数采工厂上线首月实现 30 人规模并行采集，单日有效数据生产达 100+ 小时，目前已构建 3000+ 小时单臂操作数据用于基座模型训练，预训练模型已实现覆盖业内常见 demo。
 - 主导数据存储、增强、标注业务。搭建了 PB 级本地分布式存储系统和跨平台云端（阿里云、火山云数据管理）协同，构建对象存储（OSS、TOS）和文件存储（CPFS、vePFS）的训练管理流程，完成数据存储、上云及预处理数据管线打通。
 - 主导灵巧手触觉传感器的技术调研、测试评估及供应链对接业务。测试、对比分析市面主流触觉传感器（如帕西尼、他山、墨现、华威科、华力创、视触觉传感器等）；组织供应商技术交流、方案评估及工厂调研，支撑硬件选型与系统集成决策；基于 Diffusion Policy，完成二指夹爪触觉采集系统与实机推理链路集成，实现触觉信息集成的实机操作。
2. 迅达（中国）电梯有限公司（市场部，产品管理实习生） 2017.11 – 2018.03
- 参与大区一级代理商展厅、销售方案规划和市场数据分析

获奖情况

- | | |
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| 1. HKUST ECE Future Leader | 2025 |
| 2. German Academic Exchange Service (DAAD) Alnet fellowship (AI4Science) | 2024 |
| 3. NextGen Scholar Award (IEEE Annual International Conference of EMBS) | 2024 |
| 4. 浙江大学研究生毕业奖学金 (毕业学年 Top 1%) | 2021 |
| 5. 之江杯全球人工智能竞赛（行人多目标跟踪算法开发，优胜奖 10/233） | 2019 |
| 6. 第七届全国大学生机械产品创新设计竞赛，一等奖 | 2016 |
| 7. 同济大学学业奖学金，二等、三等奖 | 2013-2018 |

- **Coding & AI:** Python / Matlab / Claude/ Opencode
- **产品设计:** AutoCAD / Inventor / Solidworks / Ansys
- **语言:** 中文、英文、德语

📄 Publications

[Working Papers]:

1. **Mingdong Li**, Zhiwei Song, Shuhang Chen, Xiang Zhang, Yiwen Wang*. Dynamic Functional Neural Connectivity Inference for Multi-Task Neuroprosthetic Control in a Point Process Filter, **IEEE Transactions on Biomedical Engineering (TBME)**. (under revision)
2. **Mingdong Li**, Jieyuan Tan, Shenghui Wu, Zhiwei Song, Mingyi Wang, Shicheng Qiu, Zixu Wang, and Yiwen Wang*. Adaptive Superposition Point Processes for Modeling Neural Population Representations Underlying Behaviors in Brain-machine Interface, **Cyborg and Bionic Systems**. (to be submitted)

[Journals]:

1. Shenghui Wu, Zhiwei Song, Xiang Zhang, Yifan Huang, Shuhang Chen, Xiang Shen, Jieyuan Tan, **Mingdong Li**, Ziyi Wang, Yujun Chen, Kai Liu, Dario Farina, Jose Principe, Yiwen Wang*. A generative spike prediction model using behavioral reinforcement for re-establishing neural functional connectivity, **Nature Computational Science**, 2026.
2. Zhiwei Song, Xiang Zhang, **Mingdong Li**, Jieyuan Tan, Yiwen Wang*. An Online Knowledge Transfer Framework for Task Learning in Brain-Machine Interfaces, **IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)**, 2025.
3. **Mingdong Li**, Shuhang Chen†, Xiang Zhang, Yiwen Wang*. Neural Correlation Integrated Adaptive Point Process Filtering on Population Spike Trains, **IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)**, 2025. (†: co-first author)
4. **Mingdong Li**, Shanhe Lou*, Hao Zheng, Yixiong Feng, Yicong Gao, Siyuan Zeng, Jianrong Tan. A Cognitive Analysis-based Key Concepts Derivation Approach for Product Design, **Expert Systems With Applications (ESWA)**, 2024.
5. **Mingdong Li**, Shanhe Lou*, Yicong Gao, Hao Zheng, Bingtao Hu, Jianrong Tan. A Cerebellar Operant Conditioning-inspired Constraint Satisfaction Approach for Product Design Concept Generation, **International Journal of Production Research (IJPR)**, 2023.
6. Xuanyu Wu, Zhaoxi Hong*, Yixiong Feng, **Mingdong Li**, Shanhe Lou, Jianrong Tan. A Semantic Analysis-driven Customer Requirements Mining Method for Product Conceptual Design, **Scientific Reports (Sci. Rep.)**, 2022.
7. Yixiong Feng, **Mingdong Li**, Shanhe Lou*, Yicong Gao, Jianrong Tan. A Digital Twin-Driven Method for Product Performance Evaluation Based on Intelligent Psycho-Physiological Analysis, **ASME Journal of Computing and Information Science in Engineering (JCISE)**, 2021.

[Conferences & Patents]:

1. **Mingdong Li**, et al. Estimating Neural Representations Underlying Behaviors via Point Process Filtering in Brain-machine Interface Task Learning, **IEEE EMBS 12th Annual International Conference on Neural Engineering (NER)** (Spotlight, top 5%)
2. **Mingdong Li**, et al. An Adaptive Superposition Point Process Model with Neuronal Encoding Engagement Identification, **2024 46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)**. (Oral)
3. **Mingdong Li**, et al. Tracking the Dynamic Functional Neural Connectivity via Conjugate Gradient Optimization, **2023 45th EMBC**. (Oral)
4. **Mingdong Li**, et al. Modeling Neural Connectivity in a Point-Process Analogue of Kalman Filter, **2022 44th EMBC**.
5. Y. Feng, **Mingdong Li**, Y. Gao. CN110090818B, 发明专利授权, 2020
6. M. Li, Y. Zhang, N. Xu, **Mingdong Li**, S. Liu. CN106127958B, 发明专利授权, 2018

项目经历

1. 针对侵入式脑机接口的点过程滤波及神经群体编码研究（博士课题） 2021-2025
2. 产品方案智能设计（硕士课题，设计文档分析，机器学习方案求解和 EEG 分析） 2018-2021
3. 智能手机 IMU + Matlab Mobile 步数测量与运动轨迹重建 2019
4. 上海市大学生创新创业计划项目（糖果包装机产品设计、样机搭建、专利书写） 2015-2016

Service

- 审稿人：
 - Journal of Neural Engineering (JNE), IEEE Transaction on Intelligent Vehicles (TIV), npj Artificial Intelligence
 - Conference: EMBC, ISBI, NER
- 课程助理：
 - EMIA4110 Practical Machine Learning 2024 Spring
 - ELEC4130 Machine Learning on Images 2023 Spring
- 会议/论坛组委会：
 - 3rd and 4th International Workshop on Neural Engineering & Rehabilitation